Agentic Statistical Analysis Assistant

CS7327 | Midterm writeup | Jingcheng Zhou and Junliang Lyu

What we built

We built a local English-language assistant for users who can state a research question but need help choosing and interpreting a statistical analysis. It accepts a CSV and a question, clarifies variable roles and study design, runs a supported test and explains the evidence. Our Python LangGraph StateGraph has 17 nodes and supports Welch independent-samples t-tests, Pearson correlation, chi-square tests of independence and simple ordinary least-squares regression. The current interfaces are a local browser, command line and notebook. Its main unresolved problem is the reliability of interpretation review.

Agent roles and why they are separate

Model A uses Qwen3:8b through Ollama to plan the analysis, assess the meaning of variables and draft the explanation. Model B uses Gemma3:12b to review that explanation against the original question, confirmed design and raw statistical evidence. These are two distinct models. Shared graph state passes the plan, diagnostics, statistics and review feedback between their roles. Model B can request a revision or withhold an explanation that remains unacceptable.

LLM checks of the research goal, variable meaning and design run in parallel with Python checks of numerical suitability; the graph then merges both branches. Python validates the plan and calculates the statistics. A deterministic check verifies numerical placeholders before Model B reviews the interpretation. Language understanding belongs in the models, while calculations and repeatable checks remain in code.

One end to end walkthrough

Our tutoring demonstration uses 80 synthetic student records, with 40 in each group. The question asks whether mean exam score differs between students with and without tutoring. The plan identifies score as the continuous outcome and tutoring as the two-group variable. Supplied teaching metadata describe independent observations and nonrandom group assignment; those facts cannot be inferred from the values alone.

The plan specifies Welch for a difference between two independent group means without requiring equal variances. Python checks usable values, group sizes, variation and distribution warnings. The hypotheses are H0: mean difference = 0 and H1: mean difference ≠ 0, with $\alpha = 0.05$. These checks cannot prove independence, and synthetic data provide a demonstration rather than evidence about actual students.

SciPy gives mean scores of 75.9525 with tutoring and 69.0350 without tutoring. The difference, tutoring minus no tutoring, is 6.9175 points, with 95% CI [2.9689, 10.8661], $t = 3.4927$, $df = 71.4775$ and $p = 0.000824$. We reject H0 under the stated assumptions. The report describes an observed association; nonrandom assignment prevents a causal claim. The graph permits one plan revision, one report revision, two answered clarification rounds and at most 15 model calls.

Follow up questions and memory

A request to explain the result again reuses stored statistics and warnings. A request for a 99% confidence interval recomputes the interval as [1.6763, 12.1587] for this example, while retaining the hypotheses and $\alpha = 0.05$. A new research question returns to planning. Date subsets currently require a newly supplied CSV; automatic date filtering is not implemented. Memory lasts only for the running graph process. Changed data require fresh diagnostics.

What basic statistical checks mean here

The workflow checks column types, missing or nonfinite values, sample sizes and variable variation. Chi-square checks expected cell frequencies; unsupported sparse tables can stop analysis. OLS includes Cook's distance, leverage and a studentized Breusch-Pagan diagnostic for unequal error variance. Flags prompt caution, not automatic removal or estimator changes. Data corrections and exclusions require explicit user confirmation; explanatory notes are optional. The original data and modification log are preserved. Paired or clustered designs and multivariable regression remain outside the current scope. A passing diagnostic does not prove that assumptions hold.

A genuine breaking scenario

In earlier live local runs, Pearson and regression produced correct statistics but their reports were withheld after the allowed revision. A literal zero in the null hypothesis conflicted with the numerical placeholder policy. This was an observed integration failure: the arithmetic succeeded, but the assistant failed to deliver an acceptable explanation.

We repaired the handling of null hypotheses, decisions and effect direction, including the distinction between a null slope of zero and an estimated slope. The latest targeted reruns delivered Pearson after one revision and regression without revision. Reviewer false alarms about rounding have also occurred. These repairs improve delivery, but disagreement between two models still does not establish which interpretation is correct.

Evaluation and what the evidence supports

At implementation snapshot 4ebe82c, all 192 software tests passed. Live evaluation on 31 predefined scenarios met adapted workflow expectations in 27 cases; all 19 assessed numerical comparisons matched their references. In a separate controlled Model B evaluation, all four valid report templates were accepted and ten of twelve planted errors were caught. The two missed errors show that an independent model does not guarantee a faithful interpretation. These are automated checks; independent human scoring is pending.

Future work

Human evaluation. Data correction and exclusion confirmations already exist. We will complete the prepared human_review.csv rubric with independent ratings for method choice, numerical fidelity and interpretation, recording misleading claims and reviewer disagreement. We may add selective analyst review for unresolved issues.

Broader methods. We will extend selection beyond the four basic methods to paired comparisons and multivariable regression, with design-specific assumptions and diagnostics. Each alternative must answer the intended statistical question; a rank test is not an automatic substitute for comparing means.

Real-world datasets. Current benchmarks use controlled synthetic data. We will add curated public datasets, such as Kaggle datasets, after checking provenance, variable definitions and research design. Analyst reference analyses will assess suitable methods and responsible interpretation. Synthetic benchmarks will remain for reproducible numerical checks.